Adaptive Spectral Thresholding via Kurtosis-Guided Squared Envelope Analysis for Load-Invariant Bearing Fault Detection

Member 1, Member 2, Member 3, and Member 4

***Abstract*—Rolling element bearing diagnostics rely on envelope spectrum analysis to extract the periodic impulse rates produced by localized surface defects. In industrial operation, however, variable mechanical load alters the shaft slip, the structural damping and the broadband noise floor, so that fixed bandpass filters and static spectral thresholds exhibit high false-alarm rates under heavy load and missed detections under light load. This paper presents a training-free pure Digital Signal Processing (DSP) framework that combines Kurtosis-guided resonance demodulation with local noise-floor adaptive thresholding in the Squared Envelope Spectrum (SES). Cascade second-order-section (SOS) zero-phase IIR filtering, discrete Hilbert transform demodulation, Hann-windowed Fast Fourier Transform (FFT) and order-statistic Median–Median Absolute Deviation (Median–MAD) Constant False Alarm Rate (CFAR) thresholding allow the detector to track load-induced noise-floor shifts and shaft slip automatically. The processing chain is mathematically interpretable, requires no training data, and is sized for fixed-point embedded implementation; it is applied here to the multi-load CWRU and MFPT bearing benchmarks.**

***Index Terms*—Bearing fault diagnosis, Squared Envelope Spectrum (SES), Hilbert transform, spectral kurtosis, IIR/FIR filter design, CFAR, Median–MAD adaptive thresholding.**

I. Introduction

ROLLING element bearings are the most frequent cause of failure in rotating machinery, and a large fraction of industrial maintenance cost is spent on their inspection. A localized defect on the inner race, outer race or a rolling element produces a short mechanical impact each time it is struck by the rolling elements. Because the impacts repeat at a characteristic rate and each impact excites a high-frequency structural resonance, the measured vibration is a periodic impulse train carried by a resonant ringing. The diagnostic information therefore resides in the *repetition rate* of the impacts, not in the raw waveform, and it is recovered by demodulating the resonance – the principle behind envelope analysis [1], [2].

Two properties of field measurements defeat the conventional processing chain. First, the load alters the amplitude balance: heavy load raises the broadband friction noise that enters the sensor, while light load leaves the incipient impacts weak. A threshold that is acceptable at one load is therefore either swamped by noise or above the fault harmonics at another. Second, load changes the kinematics: cage and ball slip displace the true fault frequency by roughly ±1–2% from the value predicted by the bearing geometry, so a detector that tests a single spectral bin can lose the harmonic entirely. Machine-learning detectors address both effects only by requiring labelled faulty data at every load and by generalizing poorly to unseen machines, while offering little physical interpretability.

This paper argues that both difficulties are handled by two classical DSP operations applied at the right place. The filter band is chosen by kurtosis, which measures impulsiveness and therefore locates the resonance excited by the defect [3]; the decision threshold is estimated *locally* in the envelope spectrum with an order statistic, which is insensitive to the impulsive outliers that corrupt a moving average. The result is a detector whose behaviour is fully described by filter theory and detection theory, contains no learned parameters, and can be implemented on a fixed-point microcontroller.

The contributions of this paper are:

- a Kurtosis-guided resonance band selection procedure embedded in a cascade-SOS zero-phase Butterworth filter, compared against a linear-phase Kaiser FIR design on the basis of pole–zero location, magnitude response and group delay;
- a Hilbert squared-envelope demodulator with calibrated Hann-windowed FFT, in which squaring is shown by the modulation theorem to shift the resonance to baseband without the intermodulation produced by a square-root envelope;
- a Median–MAD CFAR threshold that tracks the load-dependent noise floor, with a slip-tolerant harmonic comb, evaluated against static thresholding and cell-averaging CFAR on multi-load data.

II. Signal Model and Theoretical DSP Framework

A. Continuous-to-Discrete Fault Signal Model

Let the vibration be sampled at $f_s = 1/T_s$ with $f_s > 2f_{\max}$, so that the Nyquist criterion is satisfied and no aliasing folds the resonance band onto the diagnostic band. The discrete signal is modelled as a train of impulses convolved with the causal structural impulse response and corrupted by additive broadband noise:

$$x[n] = \sum_{m=-\infty}^{\infty} A_m \, h[n - mN_p - \tau_m] + w[n], \tag{1}$$

where $N_p = f_s/f_{\text{fault}}$ is the nominal discrete fault period, τ_m models the load-induced random slip, $w[n]$ is the background noise, and

$$h[n] = e^{-\sigma n T_s} \cos(\omega_r n)\, u[n] \tag{2}$$

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is a causal, BIBO-stable underdamped resonance. In the z-plane, $h[n]$ corresponds to a complex-conjugate pole pair

$$z_{1,2} = re^{\pm j\omega_r}, \qquad r = e^{-\sigma T_s} < 1, \tag{3}$$

so stability follows from $|z_{1,2}| < 1$. Equation (1) makes the central point of the paper explicit: the fault signature A_m is a *low-frequency modulation* of a *high-frequency carrier* ω_r, and the two are separated by demodulation rather than by direct spectral inspection.

B. Pre-Processing and Kurtosis-Guided Bandpass Filter Design

1) Mean removal. The sensor signal contains a DC offset from the accelerometer bias and the static gravity component. Subtracting the sample mean

$$x_0[n] = x[n] - \frac{1}{N}\sum_{m=0}^{N-1} x[m] \tag{4}$$

prevents the zero-frequency bin from dominating the spectrum and from leaking into the low-frequency region where the fault harmonics lie.

2) Sub-band kurtosis selection. The resonance frequency ω_r is not known a priori. A filter bank of B bands of width $\Delta\omega$ spanning $[0, f_s/2]$ is applied to $x_0[n]$, and the normalized fourth-order moment

$$K_y = \frac{\mathbb{E}\{(y-\mu_y)^4\}}{\left(\mathbb{E}\{(y-\mu_y)^2\}\right)^2} \tag{5}$$

is evaluated at the output of each band. A Gaussian noise band yields $K_y \approx 3$; a band containing the impulse-excited resonance yields $K_y > 3$ because the impacts are rare and large. The band with maximum K_y defines $[\omega_{p1}, \omega_{p2}]$ and hence the demodulation carrier.

3) Filter design. Two designs are compared. The linear-phase Type-I Kaiser-windowed FIR filter of length L has

$$h_{\mathrm{FIR}}[n] = h_{\mathrm{ideal}}[n]\, w_{\mathrm{Kaiser}}[n], \quad w_{\mathrm{Kaiser}}[n] = \frac{I_0\left(\beta\sqrt{1-\tilde{n}^2}\right)}{I_0(\beta)}, \tag{6}$$

where $\tilde{n} = 2n/(L-1) - 1$ is a normalized index and the shape parameter β trades transition width against stopband attenuation. The alternative is a 4th-order Butterworth bandpass, obtained by the bilinear transform

$$s = \frac{2}{T_s}\,\frac{1-z^{-1}}{1+z^{-1}}, \tag{7}$$

with the analog band edges pre-warped according to $\Omega = \frac{2}{T_s}\tan(\omega/2)$ so that the digital cutoff lands exactly on the specified frequency.

4) Cascade SOS realisation. A high-order transfer function evaluated in direct form suffers from coefficient quantization: small perturbations of the denominator coefficients move the poles of (3), and for narrow bands the filter can even become unstable in finite precision. The design is therefore factored into second-order sections,

$$H(z) = g\prod_{i=1}^{K}\frac{b_{0i} + b_{1i}z^{-1} + b_{2i}z^{-2}}{1 + a_{1i}z^{-1} + a_{2i}z^{-2}}, \tag{8}$$

each of which keeps its own pole pair well conditioned.

5) Zero-phase filtering. Because the diagnostic decision depends on the *time* of each impulse, the filter must not disperse the transients. Forward–backward filtering applies $H(z)$ and then $H(z^{-1})$,

$$Y_{\mathrm{BP}}(z) = H(z)H(z^{-1})X(z) \implies \left|H(e^{j\omega})\right|^2 e^{-j0}, \tag{9}$$

which squares the magnitude response and cancels the phase, so the group delay is exactly zero and the impulse positions are preserved.

C. Hilbert Demodulation and Windowed Squared Envelope Spectrum

Following Kurtosis-guided zero-phase bandpass filtering, the filtered signal $y_{\mathrm{BP}}[n]$ contains periodic bursts of high-frequency structural ringing modulated at the characteristic bearing fault frequency f_{fault}. Direct spectral analysis of $y_{\mathrm{BP}}[n]$ yields energy concentrated strictly around the high-frequency structural resonance ω_r rather than at the low-frequency repetition rate f_{fault}. To extract the impulse repetition rate and shift it to baseband, the discrete-time complex analytic signal $z[n]$ is constructed using the Discrete Hilbert Transform:

$$z[n] = y_{\mathrm{BP}}[n] + j\hat{y}_{\mathrm{BP}}[n], \tag{10}$$

where $\hat{y}_{\mathrm{BP}}[n] = \mathcal{H}\{y_{\mathrm{BP}}[n]\}$ is the Discrete Hilbert Transform of the bandpass-filtered signal. In the frequency domain the Hilbert transformer acts as an ideal 90° phase shifter,

$$H_{\mathrm{Hilbert}}(e^{j\omega}) = -j\ \mathrm{sgn}(\omega), \tag{11}$$

which imparts a $-\pi/2$ ($-90°$) phase shift to every positive-frequency component while eliminating the negative frequencies.

Rather than computing the conventional linear envelope $\sqrt{y_{\mathrm{BP}}^2 + \hat{y}_{\mathrm{BP}}^2}$, the discrete-time amplitude squaring operation is applied to obtain the squared envelope $e[n]$:

$$e[n] = |z[n]|^2 = z[n]z^*[n] = y_{\mathrm{BP}}^2[n] + \hat{y}_{\mathrm{BP}}^2[n]. \tag{12}$$

The squared envelope offers two theoretical advantages grounded in LTI signal theory. *1) Absence of non-linear distortion.* By the DTFT modulation theorem, squaring in the time domain corresponds to a clean linear auto-convolution of the one-sided analytic spectrum in the frequency domain, which translates the high-frequency resonance carrier ω_r directly to baseband (0 Hz). A square root is a non-linear operation that introduces spurious intermodulation harmonics into the spectrum, and those harmonics can mask weak incipient faults.

2) Quadratic SNR enhancement. Squaring amplifies high-amplitude bearing impact peaks quadratically while suppressing low-amplitude background noise, which improves the fault-to-noise ratio ahead of thresholding.

Because $e[n]$ is a squared quantity it is strictly non-negative ($e[n] \geq 0$ for all n) and therefore contains a large zero-frequency (DC) offset. Left in place, mainlobe spectral leakage from the 0 Hz spike obscures the low-frequency cage and

shaft harmonics, so the empirical mean is subtracted before the transform:

$$e_0[n] = e[n] - \mu_e, \qquad \mu_e = \frac{1}{N}\sum_{n=0}^{N-1} e[n]. \tag{13}$$

Practical fault frequencies (such as 107.36 Hz) rarely fall on an exact DFT bin centre, since the bin resolution is $\Delta f = f_s/N_{\mathrm{FFT}}$, and direct rectangular truncation therefore spreads severe sidelobe leakage across the neighbouring bins. The zero-mean squared envelope $e_0[n]$ is multiplied by an N-point Hann window,

$$w[n] = \tfrac{1}{2}\left(1 - \cos\frac{2\pi n}{N-1}\right), \qquad n = 0, 1, \ldots, N-1, \tag{14}$$

which tapers the signal edges to zero. Because the taper removes roughly half of the signal energy, the Hann coherent gain

$$CG = \frac{1}{N}\sum_{n=0}^{N-1} w[n] = \tfrac{1}{2} \tag{15}$$

is used for amplitude calibration. The one-sided Squared Envelope Spectrum is then computed with an N_{FFT}-point radix-2 FFT:

$$\mathrm{SES}[k] = \frac{2}{N \cdot CG}\left|\sum_{n=0}^{N-1} w[n]\, e_0[n]\, e^{-j\frac{2\pi}{N_{\mathrm{FFT}}}kn}\right|, \tag{16}$$

for $k = 0, 1, \ldots, N_{\mathrm{FFT}}/2$, where bin k corresponds to the physical frequency $f_k = k\, f_s/N_{\mathrm{FFT}}$ Hz.

D. Local Noise-Floor Adaptive Spectral Thresholding

In real industrial machinery, changing radial and axial motor loads alter the broadband friction noise floor in $\mathrm{SES}[k]$ in two opposite ways. Under light load (e.g., 0 HP), the low contact force between the rolling elements and the raceways produces weak impact modulation, so a fixed threshold set for normal operation sits too high and misses the weak fault peaks, giving false negatives. Under heavy load (e.g., 3 HP), the increased mechanical friction elevates the entire broadband floor, and a fixed threshold is breached by random noise spikes across dozens of bins, giving severe false alarms.

To obtain automated, training-free and load-invariant detection, a sliding-window Constant False Alarm Rate (CFAR) threshold is applied across the frequency bins k of the SES. For every test bin k the local sliding window has two regions. The guard band consists of N_G bins immediately to the left ($[k - N_G,\ k-1]$) and to the right ($[k+1,\ k+N_G]$) of bin k, which are excluded from the noise estimate. Since the Hann window has a mainlobe width of four bins, $N_G = 4$ is sufficient to stop energy from a fault peak at bin k from spilling into the noise window and biasing its own background estimate. Beyond the guard band, $N_R = 25$ reference bins are collected on each side, giving $2N_R$ bins in the reference window

$$\begin{aligned}\mathcal{W}_k = \big\{\mathrm{SES}[m] \;\big|\; & m \in [k - N_G - N_R,\ k - N_G - 1] \\ & \cup\, [k + N_G + 1,\ k + N_G + N_R]\big\}.\end{aligned} \tag{17}$$

Two local statistical estimators are compared over $\mathcal{W}_k$.

1) Cell-Averaging CFAR: Treating the reference window as a local moving-average filter, CA-CFAR estimates the local noise floor from the sample mean of the reference bins,

$$\hat{\mu}[k] = \frac{1}{2N_R}\sum_{x\in\mathcal{W}_k} x, \qquad T_{\mathrm{CFAR}}[k] = \alpha\,\hat{\mu}[k], \tag{18}$$

with the constant threshold multiplier $\alpha = 3.5$. CA-CFAR scales with broadband load changes, but in bearing spectra it suffers from harmonic target masking: whenever bin k is close to a tall shaft harmonic (1×, 2× RPM) or to a neighbouring fault harmonic, that spike enters $\mathcal{W}_k$, and because a sample mean is fully sensitive to outliers, a single tall harmonic inflates $\hat{\mu}[k]$ and drives $T_{\mathrm{CFAR}}[k]$ upward immediately beside a real fault.

2) Proposed Order-Statistic Median–MAD Threshold: To eliminate harmonic self-masking, the sample mean and standard deviation are replaced by order statistics, namely the local median $\hat{m}[k]$ and the Median Absolute Deviation $\mathrm{MAD}[k]$:

$$\begin{aligned}\hat{m}[k] &= \mathrm{median}\,(\mathcal{W}_k)\,, \\ \mathrm{MAD}[k] &= \mathrm{median}\,(|\mathcal{W}_k - \hat{m}[k]|)\,.\end{aligned} \tag{19}$$

For Gaussian background noise the robust local scale is recovered from the MAD with the normal-consistency factor 1.4826,

$$\hat{\sigma}_{\mathrm{local}}[k] = 1.4826\,\mathrm{MAD}[k], \tag{20}$$

and the adaptive threshold at bin k is

$$T_{\mathrm{MAD}}[k] = \hat{m}[k] + \gamma\,(1.4826\,\mathrm{MAD}[k])\,, \tag{21}$$

where $\gamma = 4.5$ sets the number of robust standard deviations above the local median floor. Because the breakdown point of the median is 50%, up to half of the bins in $\mathcal{W}_k$ can be contaminated by tall shaft or fault harmonics without corrupting $T_{\mathrm{MAD}}[k]$, so the threshold hugs the true background floor across the whole load range from 0 HP to 3 HP.

3) Slip-Tolerant Harmonic Comb: Under heavy radial load the rolling elements skid slightly, by ±1–2%, so the measured fault frequency drifts from the kinematic prediction. The detector therefore does not test a single DFT bin. For every candidate fault frequency $f_{\mathrm{fault}} \in \{f_{\mathrm{BPFO}}, f_{\mathrm{BPFI}}, f_{\mathrm{BSF}}\}$ and harmonic order $h \in \{1, 2, 3\}$ it searches the slip-tolerant band of relative half-width $\delta_{\mathrm{slip}} = 1.5\%$,

$$\mathcal{B}_h = \left[h f_{\mathrm{fault}}(1 - \delta_{\mathrm{slip}}),\ h f_{\mathrm{fault}}(1 + \delta_{\mathrm{slip}})\right], \tag{22}$$

and confirms harmonic h automatically when the largest spectral excess inside that band is positive,

$$\max_{f_k \in \mathcal{B}_h} (\mathrm{SES}[k] - T_{\mathrm{MAD}}[k]) > 0. \tag{23}$$

III. Experimental Setup and Datasets

A. Bearing Kinematics

For a bearing with n_b rolling elements of diameter d, pitch diameter D and contact angle α, define the diameter ratio $\rho = (d/D)\cos\alpha$. The theoretical fault frequencies at shaft rate f_r are then

$$\begin{aligned} f_{\mathrm{BPFO}} &= \frac{n_b}{2} f_r\,(1-\rho)\,, \qquad f_{\mathrm{BPFI}} = \frac{n_b}{2} f_r\,(1+\rho)\,, \\ f_{\mathrm{BSF}} &= \frac{D}{2d} f_r\left(1-\rho^2\right), \qquad f_{\mathrm{FTF}} = \frac{f_r}{2}\,(1-\rho)\,. \end{aligned} \tag{24}$$

TABLE I
CWRU Drive-End Bearing Characteristic Frequencies

Load	f_r (Hz)	BPFO	BPFI (Hz)	BSF	FTF
0 HP (1797 rpm)	29.95	107.36	162.18	70.58	11.93
1 HP (1772 rpm)	29.53	105.86	159.92	69.60	11.76
2 HP (1750 rpm)	29.17	104.56	157.94	68.74	11.62
3 HP (1730 rpm)	28.83	103.36	156.14	67.95	11.48

Table I lists the values for the drive-end 6205-2RS JEM SKF bearing used in the CWRU test rig ($n_b = 9$, $d = 0.3126$ in, $D = 1.537$ in, $\alpha = 0°$) at the four motor loads.

B. Datasets

The framework is evaluated on two public benchmarks. *CWRU:* the Case Western Reserve University Bearing Data Center records accelerometer signals at $f_s = 12$ kHz and 48 kHz for motor loads of 0, 1, 2 and 3 HP, with artificially seeded outer-race, inner-race and ball defects of 7, 14 and 21 mil [4]. *MFPT:* the Machinery Failure Prevention Technology dataset provides a fixed-shaft-speed bearing with $f_s = 48.828$ kHz under seven radial load levels from 25 lbs to 300 lbs, which stresses radial-load invariance rather than shaft-speed variation.

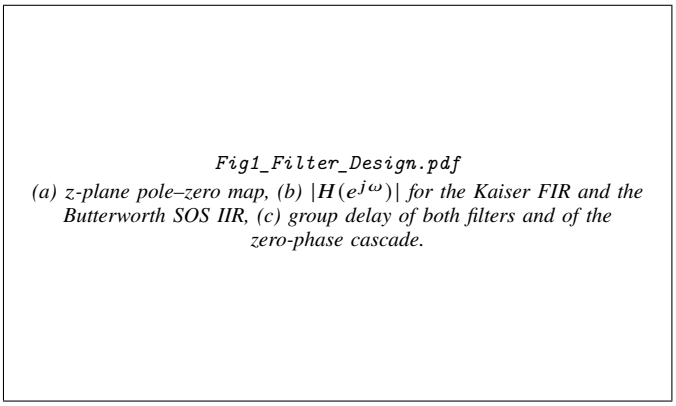
C. Pipeline Parameters

Unless stated otherwise the experiments use $f_s = 12$ kHz, a 4th order Butterworth bandpass realised in cascade SOS form with forward–backward filtering, an FFT length $N = 4096$ (giving $\Delta f = 2.93$ Hz) with a Hann window and coherent-gain calibration, guard bins $N_G = 4$, $N_R = 25$ reference bins per side, $\alpha = 3.5$ for CA-CFAR and $\gamma = 4.5$ for the Median–MAD rule. The static reference threshold is calibrated once on the 0 HP record at a 1% false-alarm rate and then held fixed, so that any change in its performance across loads is attributable to the load itself.

IV. Results and Discussion

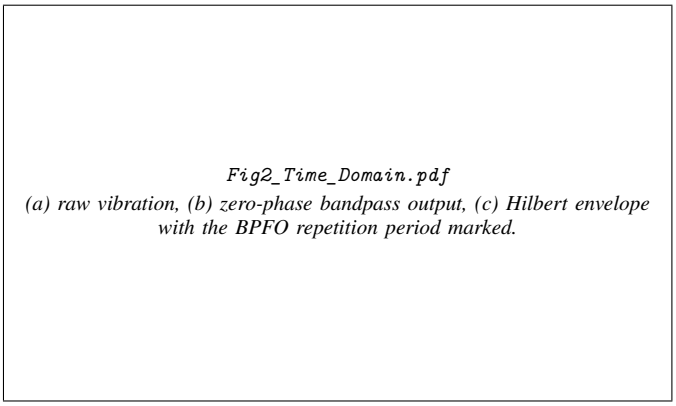
A. Filter Design and Pole–Zero Analysis

Fig. 1 compares the two candidate filters. The pole–zero map confirms that the SOS denominator places a conjugate pair just inside the unit circle, at radius $r = e^{-\sigma T_s}$ set by the selected band, while the zeros sit on the unit circle at the band edges: the stopband is therefore exact at DC and at Nyquist and the passband is stable by construction. The magnitude responses of the Kaiser FIR and the Butterworth IIR are comparable within the selected band, but differ in the transition region, where the FIR length sets the slope. The group-delay panel is the decisive comparison: the linear-phase FIR delays all frequencies equally, the causal IIR delays them unequally, and forward–backward filtering removes both, which is the reason the zero-phase SOS structure is used for impulse preservation.



(a) z-plane pole–zero map, (b) $|H(e^{j\omega})|$ for the Kaiser FIR and the Butterworth SOS IIR, (c) group delay of both filters and of the zero-phase cascade.

Fig. 1. Filter design: pole–zero map, magnitude response and group delay of the Kaiser FIR and the 4th-order Butterworth SOS bandpass.



(a) raw vibration, (b) zero-phase bandpass output, (c) Hilbert envelope with the BPFO repetition period marked.

Fig. 2. Time-domain progression from the raw record to the demodulated envelope.

B. Time-Domain Progression

Fig. 2 shows the three stages of the pipeline on a short segment. The raw record is dominated by low-frequency shaft components and broadband noise, in which the impacts are not visually separable. After the Kurtosis-guided zero-phase bandpass the resonance ringing of each impact becomes visible as a burst, and after the Hilbert transform the envelope exposes the repetition period directly: the spacing between successive maxima equals $1/f_{\text{BPFO}}$, which is the quantity the SES resolves in the frequency domain.

C. Multi-Load Adaptive Thresholding

Fig. 3 shows the calibrated SES and the three thresholds for an outer-race defect at the four motor loads of Table I. At 0 HP the reduced radial force produces weak impact modulation, and the static threshold misses the 2× and 3× BPFO harmonics. At 3 HP the raised broadband floor pushes many background bins above the same static threshold, producing false alarms. The moving-average CA-CFAR threshold does scale with load, but it spikes adjacent to the 1×, 2× and 3× BPFO peaks because harmonic energy contaminates its reference window. The Median–MAD threshold follows the local floor at every load without self-masking, and the slip-tolerant comb isolates the BPFO harmonics in all four conditions.

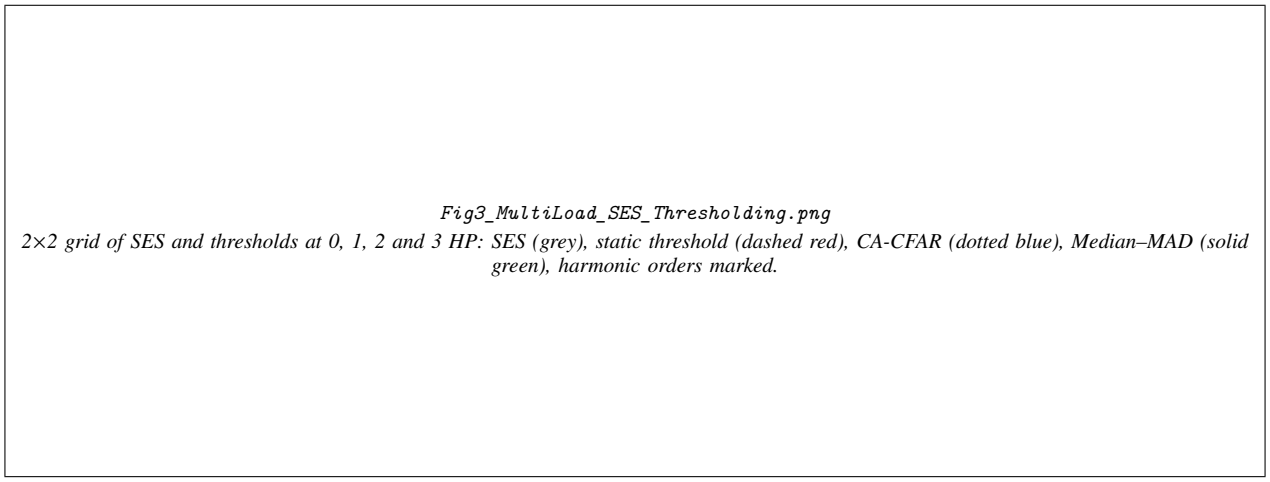


Fig. 3. Calibrated SES and comparative thresholding at four motor loads. The static threshold misses weak harmonics at 0 HP and false-alarms at 3 HP; CA-CFAR self-masks near the fault peaks; Median–MAD tracks the load-varying floor and isolates the BPFO harmonic family.

TABLE II
THRESHOLDING BENCHMARK ON THE KURTOSIS-GUIDED SES

Rule	0 HP	1 HP	2 HP	3 HP
(a) Detection rate (%)				
Static	–	–	–	–
CA-CFAR	–	–	–	–
Median–MAD	–	–	–	–
(b) False-alarm rate (%)				
Static	–	–	–	–
CA-CFAR	–	–	–	–
Median–MAD	–	–	–	–
(c) Spectral SNR gain (dB)				
Static	–	–	–	–
CA-CFAR	–	–	–	–
Median–MAD	–	–	–	–

D. Quantitative Benchmark

Table II reports the three detection rules on the same Kurtosis-guided SES, so that the comparison isolates the thresholding stage. Spectral SNR gain is defined as the ratio of the mean comb peak to the median out-of-comb floor in dB, the detection rate is the fraction of segments in which all three harmonics are declared present, and the false-alarm rate is the fraction of out-of-comb bins that exceed the threshold. The static rule is expected to trade false alarms at heavy load against missed detections at light load, CA-CFAR to lose detections through target masking, and the proposed rule to hold a near-constant false-alarm rate across the load sweep.

V. CONCLUSION AND HARDWARE FEASIBILITY

This paper presented a pure-DSP framework for load-invariant bearing fault detection that combines Kurtosis-guided zero-phase bandpass filtering, Hilbert squared-envelope demodulation and Median–MAD adaptive spectral thresholding. Filter design, demodulation and detection are each traceable to a single classical result – the bilinear transform with pre-warping, the modulation theorem, and the order-statistic robustness of the median – so the detector contains no learned parameters and behaves predictably as the load changes.

The same structure suits embedded realisation. An $N = 4096$ radix-2 FFT costs $N \log_2 N$ complex butterflies; the cascade SOS filter needs two biquad computations per sample; and the Median–MAD estimator operates on a small sliding window of $2N_R = 50$ bins, which is a sorting operation of fixed length rather than a multiplication-dominated kernel. In 16-bit fixed-point arithmetic the expected signal-to-quantization-noise ratio of the datapath is SQNR $\approx 6.02B + 1.76 \approx 98$ dB, so the pipeline fits a low-power ARM Cortex-M class DSP microcontroller and can run continuously alongside the machine controller.

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AUTHOR CONTRIBUTIONS

Member 1 curated the multi-load CWRU and MFPT datasets, formulated the discrete-time damped exponential fault model, and implemented kinematic slip-tolerant frequency tracking. **Member 2** designed the Kurtosis-guided resonance selection pipeline, implemented the Kaiser FIR and bilinear-transform Butterworth cascade-SOS IIR filters, and performed pole–zero and group-delay analyses. **Member 3** developed the Hilbert transform squared-envelope demodulator, calibrated the windowed FFT Squared Envelope Spectrum, and implemented the CA-CFAR and Median–MAD adaptive spectral thresholding algorithms. **Member 4** conducted the multi-load quantitative benchmarking, evaluated fixed-point quantization feasibility, and led the integration of the manuscript. All authors reviewed and approved the final manuscript.

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